School of Computer and Systems Sciences
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23 October 2026

The Editors
Intelligent Oncology

Dear Editors,

The authors submit the manuscript "Patient-level differential privacy, membership leakage and explanation fidelity in federated brain tumour MRI classification" for consideration as a full-length research article.

Multi-hospital artificial intelligence for neuro-oncology needs formal privacy guarantees, yet published federated brain tumour classifiers rarely state an accounted budget or the protected unit, and their public benchmarks overlap. The manuscript measures what accounted differential privacy costs and what it protects. On a deduplicated three-source benchmark, a mirror-aware check found 66 test images with near-copies in other splits that perceptual hashing had missed. Record-level differential privacy at $\varepsilon = 8$ cost 0.076 in balanced accuracy for federated and 0.130 for central training, mainly through meningioma and no-tumour sensitivity, whereas a per-patient guarantee needed $\varepsilon = 64$. With non-members matched to members by source and class, a likelihood-ratio membership attack found 11.1% of training images at a 1% false-positive rate without privacy and 1.0% to 1.5% under record-level privacy, not significantly above chance. An unclipped noise mechanism used in prior work still exposed training patients (area under the curve 0.745). Grad-CAM localisation and cross-seed consistency declined in step with accuracy.

The work fits the journal's scope in artificial intelligence for oncology imaging and builds on themes the journal has published. Ahmed et al. reviewed artificial intelligence for brain tumour detection, grading and response evaluation (Intell Oncol 2026;2(3):100071, doi:10.1016/j.intonc.2026.100071), and Jung et al. applied deep learning to small brain metastases (Intell Oncol 2026;2(2):100049, doi:10.1016/j.intonc.2026.100049). Li et al. reviewed cryptographic privacy-preserving medical image processing (Intell Oncol 2026;2(4):100083, doi:10.1016/j.intonc.2026.100083), and Cai discussed transparency and privacy preservation in medical artificial intelligence (Intell Oncol 2025;1(2):172-175, doi:10.1016/j.intonc.2025.03.006). The manuscript complements these papers with a measured privacy, utility and explanation audit for federated neuro-oncology models.

The manuscript is original, has not been published, and is not under consideration elsewhere. All authors approved the submission. The authors declare no competing interests, and the work received no specific funding. The study used only public, de-identified images. Code, frozen results and figure data are available at `https://github.com/shashiranjan-jnu/dpfl-braintumor`, archived on Zenodo under doi:10.5281/zenodo.23601090.

Suggested reviewers, with no collaboration with the authors in the past five years:

Name	Affiliation	Institutional e-mail	Expertise
1			
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Yours sincerely,

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On behalf of Shashi Ranjan and Poonguzhali N